

# Existence of Non-Equivariant Gromov's Tori

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# Result

**Theorem (R. Bryant, 1985).**  $\mathbb{T}^2$  is a flat torus isometrically and minimally embedded in  $\mathbb{S}^N$ . Then it is equivariant.

**Theorem 1.**  $n \geq 3$ . Then  $\exists \mathbb{T}^n$ , a flat torus isometrically and minimally embedded in  $\mathbb{S}^N$ , which is not equivariant.

**Theorem (M. Gromov, 2022).**  $n \geq 2, N \gg n$ . Then

$\exists \mathbb{T}^n \subset \mathbb{T}_{\text{Cl}}^N \subset \mathbb{S}^{2N-1}$ , where  $\mathbb{T}^n$  is a geodesic subtorus,  $k_p(v) \equiv \sqrt{\frac{3n}{n+2}}$  for  $p \in \mathbb{T}$ , where  $k_p(v)$  is the normal curvature. Also,  $\mathbb{T}^n$  is equivariant.

**Corollary (M. Gromov, 2022).**  $(L, g)$  is a Riemannian manifold. Then  $\exists \lambda > 0 : (L, \lambda^2 g) \hookrightarrow \mathbb{B}^q$ ,  $k_p(v) \leq \sqrt{3}$  for  $p \in L$ .

**Theorem (A. Petrunin, 2023).**  $\mathbb{T}^n \looparrowright \mathbb{B}^q$ . Then

$$\max\{k_p(v) \mid p \in \mathbb{T}, v \in T_p \mathbb{T}\} \geq \sqrt{\frac{3n}{n+2}}.$$

# Result

**Definition.** Let  $\mathbb{T}^n \looparrowright \mathbb{B}^q$  and  $k_p(v) \leq \sqrt{\frac{3n}{n+2}}$  for  $p \in \mathbb{T}$ . Then  $\mathbb{T}$  is a **Gromov torus**.

**Theorem 2.** Gromov's tori are flat, embedded in a sphere and minimal in it, and also  $k_p(v) \equiv \sqrt{\frac{3n}{n+2}}$ .

**Theorem 3.** For any  $n \geq 3$ , there exists a non-equivariant Gromov torus  $\mathbb{T}^n$ .

# Outline

1) Gromov's tori are flat and minimal in a sphere, and also

$$k_p(v) \equiv \sqrt{\frac{3n}{n+2}}.$$

2)  $f : \mathbb{R}^n \rightarrow \mathbb{C}^{q+1}$  defines a Gromov torus with lattice  $A \iff$

$$\text{Im} f \subset \mathbb{R}^{q+1}, \Delta f + nf = 0, |f| \equiv 1, |D_u f| \equiv 1, |D_u^2 f| \equiv \sqrt{\frac{3n}{n+2}}$$

$$(u \in \mathbb{S}^{n-1}), f(x+a) = f(x) \ (a \in A).$$

3.a)  $f : \mathbb{R}^n \rightarrow \mathbb{C}^{q+1}$  defines a Gromov torus with lattice  $A \iff$

$f(x) = \sum_k c_k \cdot \exp(2\pi i \langle k, x \rangle)$  for  $k \in \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1} \cap A^*$  and the conditions of item 2 are satisfied in terms of  $\{c_k\}$ .

3.b)  $f$  defines an equivariant torus  $\iff c_k \perp c_l$ .

4) Construction of a counterexample using item 3.

# Basic Concepts

$L^n \looparrowright \mathbb{R}^q$ .  $\Pi_x : T_x L \rightarrow N_x L$ ,  $\Pi_x(v, v) := \gamma_v''(0)$ , where  $\gamma_v$  is a geodesic,  $\gamma_v(0) = x$ ,  $\gamma_v'(0) = v$ .  $\Pi$  is the **second fundamental form**.

$k_x(v) := \Pi(v, v)$  is the **normal curvature**.

$H(x) := \sum_{i=1}^n k_x(e_i)$  is the **mean curvature**.

A submanifold is **minimal** if  $H(x) \equiv 0$ .

$\text{Sc}(x) := \sum_{i \neq j} \text{Sec}(e_i, e_j)$  is the **scalar curvature**.

$$\mathcal{K}(x) := \frac{1}{|\mathbb{S}^{n-1}|} \int_{|v|=1} |\Pi_x(v, v)|^2 dv.$$

# Proof of Theorem 2

## Item 1)

**Theorem (Gauss Formula).**  $Sc = \frac{3}{2}|H|^2 - \frac{n \cdot (n+2)}{2} \cdot \mathbb{K}$

**Theorem.**  $g$  is a Riemannian metric on  $\mathbb{T}^n$ ,  $n \geq 3$ ,  $Sc$  is its scalar curvature,  $u \in C^\infty(\mathbb{T}, \mathbb{R}_+)$ . Then  $(Sc \cdot u - 4 \cdot \frac{n-1}{n-2} \cdot \Delta u) \cdot u^{\frac{n-2}{n+2}}$  is the scalar curvature for the metric  $u^{\frac{4}{n-2}} \cdot g$ .

**Theorem.**  $\mathbb{T}^n$  is a torus with a Riemannian metric.

- 1)  $\exists x : Sc(x) \leq 0$ .
- 2)  $Sc(x) \geq 0$  for  $x \in \mathbb{T} \implies \mathbb{T}$  is flat.

## Proof of Theorem 2

Using the theorems above, A. Petrunin proves the following inequalities in his paper:

**Theorem.** Let  $L$  be compact.

- 1)  $L \looparrowright \mathbb{B}^q$ . Then  $\frac{1}{|L|} \int_L |H(x)| dx \geq n$ .
- 2)  $L \looparrowright \mathbb{S}^q$ . Then  $|H(x)| \geq n$  for  $x \in L$ .

**Theorem.**  $g$  is a Riemannian metric on  $\mathbb{T}^n$ ,  $n \geq 3$ ,  $Sc$  is its scalar curvature,  $u \in C^\infty(\mathbb{T}, \mathbb{R}_+)$ . Then  $\exists x \in \mathbb{T}$ :  
 $Sc(x) \cdot u(x) - 4 \cdot \frac{n-1}{n-2} \cdot \Delta u(x) \leq 0$ .

## Proof of Theorem 2

We investigate the case of equality and obtain:

**Lemma 1.**  $N^m \looparrowright \mathbb{S}^{n-1} \subset \mathbb{R}^n$ . Then  $|H_{N \subset \mathbb{R}^n}(p)| = m \implies |H_{N \subset \mathbb{S}^{n-1}}(p)| = 0$ .

**Lemma 2.**  $f : \mathbb{R}^n \rightarrow \mathbb{S}^q$  is a minimal isometric immersion. Then  $\Delta f + nf = 0$ .

As well as the proof of Theorem 2.



# Defining a Gromov Torus

## Item 2)

Let  $\mathbb{T}^n \hookrightarrow \mathbb{S}^q \subset \mathbb{R}^{q+1}$  be a Gromov torus.  $\mathbb{T}^n \cong \mathbb{R}^n / A\mathbb{Z}^n$ . Thus, a Gromov torus is defined by a map  $f : \mathbb{R}^n \rightarrow \mathbb{R}^{q+1}$  with the following properties:

- 1)  $|f| \equiv 1$ ,
- 2)  $|D_u f| \equiv 1 \ (u \in \mathbb{S}^{n-1})$ ,
- 3)  $|D_u^2 f| \equiv \sqrt{\frac{3n}{n+2}} \ (u \in \mathbb{S}^{n-1})$ ,
- 4)  $f(x+a) = f(x) \ (a \in A)$ ,
- 5)  $\Delta f + nf = 0$ .

In fact, these requirements on  $f$  are also sufficient:

**Proposition 1.**  $f : \mathbb{R}^n \rightarrow \mathbb{R}^{q+1}$  defines a Gromov torus with lattice  $A \iff$  conditions 1) – 5) are satisfied.

# Defining a Gromov Torus

## Item 3.a)

For further work, we reformulate our criterion in a more convenient form.

**Lemma 3.**  $f(x+a) = f(x)$  ( $a \in A$ ),  $\Delta f + nf = 0 \iff$   
 $f(x) = \sum_k c_k \cdot \exp(2\pi i \langle k, x \rangle)$  for  $k \in \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1} \cap A^*$ .

A map  $f$  defining a Gromov torus satisfies the conditions of Lemma 3. Then, from Proposition 1 and Lemma 3:

**Proposition 2.**  $f : \mathbb{R}^n \rightarrow \mathbb{C}^{q+1}$  defines a Gromov torus with lattice  $A \iff f(x) = \sum_k c_k \cdot \exp(2\pi i \langle k, x \rangle)$  for  $k \in \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1} \cap A^*$ ,  
 $\text{Im} f \subset \mathbb{R}^{q+1}$ ,  $|f| \equiv 1$ ,  $|D_u f| \equiv 1$ ,  $|D_u^2 f| \equiv \sqrt{\frac{3n}{n+2}}$  ( $u \in \mathbb{S}^{n-1}$ ).

# Criterion in Terms of Coefficients

1.

$$\sum_{j \in M} |c_j|^2 = 1;$$

2.

$$\sum_{j \in M} |c_j|^2 (k_j^\alpha)^2 = \frac{1}{4\pi^2};$$

3.

$$\sum_{j \in M} |c_j|^2 k_j^\alpha k_j^\beta = 0, \alpha \neq \beta;$$

4.

$$\sum_{j \in M} |c_j|^2 (k_j^\alpha)^4 = \frac{3n}{16\pi^4(n+2)};$$

5.

$$\sum_{j \in M} |c_j|^2 (k_j^\alpha + k_j^\beta)^4 = \frac{3n}{4\pi^4(n+2)}.$$

# Criterion in Terms of Coefficients

$$L_v := \{(i, j) \in M \times M \mid k_i - k_j = v\}, v \in \mathbb{R}^n, 2l_v := |L_v|.$$

1.

$$\sum_{(i,j) \in L_v} \langle c_i, c_j \rangle = 0;$$

2.

$$\sum_{(i,j) \in L_v} \langle c_i, c_j \rangle k_i^\alpha k_j^\beta = 0;$$

3.

$$\sum_{(i,j) \in L_v} \langle c_i, c_j \rangle (k_i^\alpha + k_i^\beta)^2 (k_j^\alpha + k_j^\beta)^2 = 0;$$

4.

$$\langle c_i, c_j \rangle = \overline{\langle c_j, c_i \rangle};$$

5.

$$\langle c_i, c_j \rangle = \langle c_{-j}, c_{-i} \rangle.$$

# Equivariance Criterion

## Item 3.b)

If we want to define a non-equivariant Gromov torus by a map  $f$ , we will need an equivariance criterion in terms of  $\{c_k\}$ .

**Lemma 4.** Let  $f(x) = \sum_k c_k \cdot \exp(2\pi i \langle k, x \rangle)$  define a Gromov torus. Then the torus is equivariant  $\iff c_k \perp c_l$  for  $k \neq l$ .

### Proof Outline.

- 1) Torus is equivariant  $\Rightarrow$  torus is an orbit of a torus in  $O(q)$ .
- 2) Torus is an orbit of a torus in  $O(q) \Rightarrow$  Torus is equivariant.
- 3) Torus is an orbit of a torus in  $O(q) \Rightarrow c_k \perp c_l$  for  $k \neq l$ .
- 4)  $c_k \perp c_l$  for  $k \neq l \Rightarrow$  torus is an orbit of a torus in  $O(q)$ .

**A toral subgroup** of a Lie group is any connected, compact, Abelian Lie subgroup.

# Construction Outline

## Item 4)

**Step 1)** Consider an equivariant Gromov torus. We add variables to its inhomogeneous system for  $\{|c_j|^2\}$  so that the system has maximal rank and a positive solution.

**Step 2)** We straighten the obtained set of vectors while preserving the positive solution and complete it to a square lattice, so that some homogeneous system has a non-trivial solution.

**Step 3)** Using the obtained inner products, we find the coefficients  $\{c_j\}$  themselves.

# Notation

## Step 1)

Let  $\mathbb{T}^n$  be an equivariant Gromov torus with lattice  $A$ ,

$$\{k_j\}_{j=1}^m := \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1} \cap A^*.$$

$$\Phi(k) := (1, \dots, (k^j)^2, \dots, k^j \cdot k^i, \dots, (k^j)^4, \dots, (k^j + k^i)^4, \dots).$$

$$b := \left( \frac{1}{2}, \dots, \frac{1}{8\pi^2}, \dots, 0, \dots, \frac{3n}{32\pi^4(n+2)}, \dots, \frac{3n}{8\pi^4(n+2)}, \dots \right).$$

Then the inhomogeneous system with respect to  $\{|c_j|^2\}$ ,  
 $\sum_{j=1}^m \Phi(k_j) \cdot |c_j|^2 = b$ , has a non-negative solution.

$$\{k_j\}_{j=1}^r := \{k_j : |c_j|^2 > 0\}.$$

# Construction

$$V := \text{Span}(\text{Im} \Phi|_{\frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1}}), \quad V_0 := \text{Span}(\Phi(k_1), \dots, \Phi(k_r)).$$

**Lemma 5.**  $V \neq V_0$ , let  $\lambda_{\mathbb{S}}^{n-1}$  be the Lebesgue measure on  $\frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1}$ , and  $\mu$  be the left-invariant Haar measure on  $O(n)$ .

1)  $\lambda_{\mathbb{S}}^{n-1}(\Phi^{-1}(V_0)) = 0$ .

2)  $\mu(\{\varphi \in O(n) : \Phi(\varphi(k_j)) \notin V_0, j = 1, \dots, m\}) = \mu(O(n))$ .

Then, by Lemma 5, we can complete the lattice to the required one:

**Corollary.**  $\exists \{\kappa_i\}_{i=1}^{\rho} \subset \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1}$ :  $V = V_0 + \text{Span}(\Phi(\kappa_1), \dots, \Phi(\kappa_{\rho}))$  and  $\sum_{j=1}^r \Phi(k_j) \cdot |c_j|^2 + \sum_{i=1}^{\rho} \Phi(\kappa_i) \cdot |c_i|^2 = b$  has a positive solution with respect to  $\{|c_j|^2\}$ .



# Construction

## Step 2)

$B$  is a **square** lattice if its basis is proportional to an orthonormal one.

Using the corollary, we can perform the second step in the following sense:

**Lemma 6.**  $\varepsilon_A > 0$ . Then  $\exists B$ , an arbitrarily small square lattice, with  $\text{dist}(k_j, \frac{\sqrt{n}}{2\pi} S^{n-1} \cap B) < \varepsilon_A$  ( $j = 1, \dots, r$ ):

- 1)  $B^*$  defines a Gromov torus with non-zero coefficients  $\{c_j\}$ .
- 2)  $\exists \{a_j\}, \{b_{ji}\} \subset \mathbb{C}$  is a solution to the entire system for the lattice  $B^*$ ,  $\forall a_j > 0, \exists b_{ji} \neq 0$ .

# Construction

## Step 3)

Due to the homogeneity of the subsystems of interest, it suffices to prove the following lemma:

**Lemma 7.** Let  $\{a_j\}, \{b_{ji}\} \subset \mathbb{R}, \forall a_j > 0$ . Then  
 $\exists N \in \mathbb{N}, \varepsilon > 0, \{v_j\} \subset \mathbb{R}^N: |v_j| = a_j, \langle v_j, v_i \rangle = \varepsilon b_{ji}.$

Lemmas 6.2 and 7 together complete our construction and prove Theorem 3.